

A Reciprocal Local-Count Gate for the Small-Noise Limit

Rouche Stability versus Proliferating Fredholm Zeros

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Abstract

At every fixed positive noise, the Hilbert–Schmidt determinant exists and its zeros are reciprocal resonances. A locally uniform small-noise limit requires more: zero counts on any contour avoiding the limiting divisor must eventually stabilize.

We formulate this requirement as a Rouche local-count gate and audit the rank-growing reciprocal clouds on the predeclared radii 1.2, 1.5, 2, 3, 5. All 160 endpoint-contour pairs have positive finite clearance; the minimum is 1.3338×10^{-4} . Nevertheless none of the five left-channel count sequences and only one right-channel sequence is constant over the last four scales. The largest first-to-last count growth is 18. On radius two, both channels grow from two to ten selected reciprocal zeros.

This is a finite gate failure, not an asymptotic no-go theorem. The selected cloud is not the complete determinant divisor, and additional smaller-noise levels could eventually stabilize. The result says that local uniformity is not yet supported and that moving-cloud or relative-determinant renormalization cannot be skipped.

1 A necessary condition for local uniformity

Let $\Omega \subset \mathbb{C}$ be a domain and let f_n, f be holomorphic on Ω .

Theorem 1.1 (Contour-count stability). *Suppose $f_n \rightarrow f$ locally uniformly and $f \not\equiv 0$. Let $\Gamma \subset \Omega$ be a positively oriented simple closed contour on which f has no zero. Then, for all sufficiently large n , f_n and f have the same number of zeros inside Γ , counted with multiplicity.*

Proof. Compactness of Γ and nonvanishing of f give $\min_{\Gamma} |f| > 0$. Local uniform convergence eventually gives $\max_{\Gamma} |f_n - f| < \min_{\Gamma} |f|$. Rouché’s theorem then proves the claim [1]. $\square$

Thus every nonzero locally uniform determinant limit has eventual local divisor stability away from its boundary zeros. This condition is necessary, not sufficient.

2 Finite reciprocal clouds

For each selected resonance cloud $\Lambda_{\sigma,s}$ from RH-222, define

$$Z_{\sigma,s} = \{\lambda^{-1} : \lambda \in \Lambda_{\sigma,s}\}.$$

RH-226 proves that these are exactly the zeros of the selected regularized Fredholm factor [2]. For radius R , define

$$N_{\sigma,s}(R) = \#\{z \in Z_{\sigma,s} : |z| < R\} \tag{1}$$

with multiplicity, and finite contour clearance

$$c_{\sigma,s}(R) = \min_{z \in Z_{\sigma,s}} ||z| - R|. \tag{2}$$

Positive $c_{\sigma,s}(R)$ makes each individual finite count unambiguous. It does not give a uniform lower bound or identify the zero count of the unresolved operator complement.

3 Predeclared gate

The radii

$$\mathcal{R} = \{1.2, 1.5, 2, 3, 5\}$$

probe, respectively, the first near-unit reciprocal zeros and progressively larger parts of the selected divisor. The finite diagnostic asks whether each count is constant over the last four frozen noise levels

$$\sigma = 0.0025, 0.002, 0.0016, 0.00125.$$

Four levels are a diagnostic window, not a definition of eventuality.

The complete count sequences are:

side	R	first eight levels	last eight levels
left	1.2	0, 0, 0, 0, 0, 0, 0, 0	2, 2, 2, 2, 2, 2, 4, 4
left	2.0	2, 4, 4, 4, 4, 4, 6, 6	6, 6, 8, 8, 8, 8, 8, 10
left	5.0	4, 6, 7, 10, 11, 11, 12, 13	14, 17, 16, 20, 20, 19, 23, 22
right	1.2	0, 0, 0, 0, 0, 0, 0, 0	2, 2, 2, 2, 2, 2, 2, 4
right	2.0	2, 4, 4, 4, 4, 4, 6, 6	6, 6, 8, 8, 8, 8, 8, 10
right	5.0	4, 6, 7, 9, 10, 13, 12, 15	14, 17, 16, 18, 20, 19, 23, 22

The nonmonotonicity at larger radii is allowed: changing σ moves old zeros as well as adding selected shells. Rank growth alone does not make fixed-disk counts monotone.

4 Audit verdict

diagnostic	result
endpoint-contour cases	160
minimum finite contour clearance	1.3338×10^{-4}
largest first-to-last count growth	18
left radii constant on last four levels	0/5
right radii constant on last four levels	1/5
all-radius finite gate	failed

The single stable right sequence is $R = 1.5$, with final counts 8, 8, 8, 8. It does not compensate for the failures at the other radii.

At $R = 2$, both channels move from count two to count ten. If this growth continued without bound while the selected clouds captured all reciprocal zeros in the disk, it would obstruct a nonzero locally uniform limit by Theorem 1.1. The finite data do not establish either premise.

5 Why this is not a disproof

There are four logical gaps between the audit and an asymptotic no-go theorem:

1. only sixteen noise levels are observed;
2. the rank schedule is a stress test, not a complete spectral cutoff;

3. Arnoldi roots and contour clearances are not interval certified;
4. a relative determinant may remove a growing near-unit factor before taking a limit.

Therefore the exact conclusion is that the current finite family does not pass a necessary diagnostic. We do not claim that eventual stability is false.

RH-80 gives an abstract moving-cloud relative determinant route [3]. To use it here one must construct a reducing cloud and bound the complementary determinant uniformly. Before attempting that harder operator step, the next paper asks a narrower question: are the complete shells already resolved beyond the selected cloud controllable as an omitted regularized-product tail?

6 Claim boundary

Theorem 1.1 is exact. The five-radius verdict is finite. No small-noise determinant limit is disproved, and no new determinant is constructed. Gates A–E remain open. In particular the reciprocal Markov zeros are not asserted to be self-adjoint energies or zeta zeros.

References

- [1] John B. Conway. *Functions of One Complex Variable*. Springer, 1978.
- [2] Liang Wang. The reciprocal resonance–fredholm dictionary. 2026. RH-226 preprint.
- [3] Liang Wang. Moving-cloud renormalization and the relative determinant gate. 2026. RH-80 preprint.